

Title:Fermison: A State-Centric Framework for Timeless, Loss-Tolerant Quantum States

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Abstract

We propose a theoretical framework called Fermison, a quantum state defined by the combination of two sectors: a 6-state W-sector and a 2-state GHZ-sector, giving a total of 8 basis states. Fermison is state-centric: it does not rely on the Pauli exclusion principle, nor does it require spin, direction, or linear time. Instead, it defines “Alone states” (e.g., 110, 01111, 1000) as self-contained configurations that coexist harmoniously. These states form “Buddies” pairs that persist without external enforcement. Fermison exhibits loss tolerance, self-healing, measurement-free operation, and zero net heat. Its dynamics follow a natural Fibonacci rhythm, leading to golden-ratio stability and anti-resonance. Bifurcations do not lead to chaos but to ordered growth. We further speculate that dark matter may correspond to the irrational remainder of the Fibonacci spiral. This framework challenges conventional quantum assumptions and offers a new perspective on timeless, directionless quantum states.

1. Introduction

In standard quantum mechanics, the Pauli exclusion principle forbids identical fermions from occupying the same state, forcing them into separate spin configurations. This principle, while successful, may not be fundamental for all quantum systems. The Fermison framework begins with a simple question: what if particles could coexist as self-contained “Alone states” without being forced apart? We look at states, not masks. We focus on the intrinsic existence and relations of states, rather than on external rules or mathematical formalism.

2. Definitions

- W-Sector: A set of 6 basis states, denoted $|W_1\rangle$ through $|W_6\rangle$. These represent configurations that can tolerate the loss of one component while remaining functional.
- GHZ-Sector: A set of 2 basis states, $|G_1\rangle$ and $|G_2\rangle$, analogous to GHZ entangled states (e.g., $|000\rangle$ and $|111\rangle$). These preserve entanglement.
- Fermison State: The combined system of W-Sector and GHZ-Sector, spanning an 8-dimensional Hilbert space. Symbolically: $\text{Fermison} = W + \text{GHZ} = 8 \text{ states}$.
- Alone State: A self-contained quantum state that can exist independently, without obeying the Pauli exclusion principle. Examples include 110, 01111, and 1000. These states are complete in themselves.
- Buddy: A pair of states that coexist harmoniously, not because of an external rule, but because they naturally fit together. Buddies are the true paired states—not mathematical even numbers.

3. Properties

- Loss Tolerance: If one state in the W-Sector is lost, the remaining states can rearrange themselves to maintain overall stability. For example, 110 may become 11110.
- Self-Healing: The system naturally returns to a steady state through a healing operator, without external intervention.
- Measurement-Free: No measurement or observer is required for the system to function or recover.
- Zero Heat: The total energy and momentum of the system are zero; all changes cancel out. $\Delta Q = 0$.
- Symmetry Protection: The system exhibits an underlying symmetry that protects it from perturbations. A possible connection to exceptional Lie groups such as $E_8 \times E_8$ is left for future investigation.

4. Time and Force

We reject linear time as a fundamental property. Instead, we adopt Fibonacci Time: a natural rhythm (1, 1, 2, 3, 5, 8, ...) that is non-periodic, anti-resonant, and converges to the golden ratio ϕ . This rhythm governs the system's evolution and ensures ultimate stability.

Force is described through the Fibonacci Spiral, which can expand and contract at any scale. Expansion corresponds to repulsion; contraction corresponds to attraction. Because the spiral is scale-invariant and self-similar, force emerges naturally from geometry, without distance dependence or external agents. Fibonacci Time and Fibonacci Spiral are two sides of the same coin: one temporal, one spatial.

5. Chaos and Bifurcation

The system exhibits bifurcations sudden changes in behavior but these do not lead to chaos. Instead, they follow the Fibonacci rhythm and converge to the golden ratio. This is "ordered chaos": growth through branching, where local disturbances (small losses) preserve global symmetry. The system does not collapse; it adapts and becomes more robust.

6. Dark Matter as a Fibonacci Fraction

We speculate that dark matter corresponds to the irrational remainder of the Fibonacci spiral. As the spiral expands and contracts, some portions have rational ratios (visible matter), while others have irrational ratios (dark matter). Dark matter is not a new particle or force; it is simply the leftover fraction of the same geometric process. Visible and dark matter are two sides of the same Fibonacci spiral.

7. Conclusion

Fermion is a state-centric, timeless, directionless framework that challenges the Pauli exclusion principle and linear time. It defines quantum states through existence and relation, not through external rules. Its properties loss tolerance, self-healing, measurement-free operation, zero heat emerge from a Fibonacci rhythm and golden-ratio stability. Bifurcations lead to ordered growth, and dark matter appears as a natural remainder. Fermion offers a new way to think about quantum systems: not as particles obeying laws, but as states existing in harmony.

Author's Note:

"I don't use real time. I use Fibonacci time. My quantum states don't evolve second by second they evolve through bifurcation rhythms. This is how I bridge Einstein's relativity with

quantum chaos. The variable is just a mask. The state is the invariant reality that exists outside of real time."